

Chapter 2: Review of Related Literature

INSERT nato here short introductory paragraph explaining the chapter's coverage and its connection to evaluating road-network connectivity for SUHI estimation.

2.1 Related Literature

This section synthesizes the concepts, findings, and methodological evidence supporting your study. Organize studies by the questions they address, integrating Philippine research where relevant.

2.1.1 Surface Urban Heat Islands and Urban Thermal Conditions

- **2.1.1.1 Urban heat islands and surface urban heat islands**
- **2.1.1.2 Land surface temperature, air temperature, and SUHI intensity**
- **2.1.1.3 Urbanization and spatial variation in surface temperature**
- **2.1.1.4 Urban thermal conditions in tropical cities and Davao City**

Explain the phenomenon and establish exactly what the thesis estimates. Keep the regional and local discussion focused on surface thermal conditions; the broader climate and public-health motivation is already developed in Chapter 1.

Connection to the study: Establishes the meaning and boundaries of the response variable.

2.1.2 Measurement and Interpretation of SUHI Intensity

- **2.1.2.1 Urban–non-urban surface-temperature contrasts**
- **2.1.2.2 Urban boundaries and non-urban reference-area selection**
- **2.1.2.3 Reference land cover, elevation, and water exclusion**
- **2.1.2.4 Temporal matching and sensitivity to reference definitions**

Review how studies define and calculate SUHI, particularly how the reference area affects interpretation. This section should support your proposed use of a common target-year non-urban reference.

Connection to the study: Justifies the step from estimated LST to derived SUHI intensity.

2.1.3 Satellite Remote Sensing of Urban Surface Temperature

- **2.1.3.1 Thermal infrared observations and satellite-derived LST**
- **2.1.3.2 Landsat 8 surface temperature and surface reflectance products**
- **2.1.3.3 Cloud screening, missing observations, and annual compositing**
- **2.1.3.4 Spatial resolution, mixed pixels, and observation uncertainty**

Review the evidence supporting satellite-based thermal analysis and the limitations that affect your measurements. Discuss alternative sensors only briefly when useful for explaining trade-offs.

Connection to the study: Supports the Landsat 8 annual-composite workflow specified in Chapter 1.

2.1.4 Environmental Predictors of Land Surface Temperature

- **2.1.4.1 Vegetation characteristics and NDVI**
- **2.1.4.2 Built-up characteristics and NDBI**
- **2.1.4.3 Historical LST as a predictor of subsequent thermal conditions**
- **2.1.4.4 Interactions, spatial variability, and limitations of spectral indicators**

Synthesize findings about your actual predictors: historical LST, NDVI, and NDBI. Other variables, such as elevation or urban geometry, can explain omitted influences without becoming additional proposed model inputs.

Connection to the study: Establishes why the selected node features are relevant.

2.1.5 Road Networks as Spatial Representations of Urban Environments

- **2.1.5.1 Urban morphology and road-network connectivity**
- **2.1.5.2 OpenStreetMap data quality and street-network extraction**
- **2.1.5.3 Intersections, intermediate road points, and graph adjacency**
- **2.1.5.4 Road-connected neighborhoods and geographic proximity**

Examine why road graphs might provide useful relational information and what they leave out. Maintain the distinction established in Chapter 1 between connectivity as a modeling relationship and physical heat transfer.

Connection to the study: Develops the rationale for the representation whose contribution you will test.

2.1.6 Integration of Satellite Observations with Road-Network Nodes

- **2.1.6.1 Raster–vector alignment and spatial feature extraction**
- **2.1.6.2 Point sampling, buffers, and neighborhood aggregation**
- **2.1.6.3 Node densification, overlapping buffers, and shared pixels**
- **2.1.6.4 Missing-data treatment and preservation of observation quality**

Review how environmental observations are assigned to graph nodes. This deserves its own section because our earlier discussions identified important differences between buffer measurements, road-segment summaries, and map symbols.

Connection to the study: Explains what a node’s thermal value represents and why denser road points do not automatically provide finer independent measurements.

2.1.7 Machine Learning for Urban Thermal Estimation

- **2.1.7.1 Statistical and conventional machine-learning approaches**
- **2.1.7.2 Random Forest regression for thermal estimation**
- **2.1.7.3 Raster-based deep learning and spatial context**
- **2.1.7.4 Predictor selection, model comparison, and generalization**

Compare the capabilities and limitations of established approaches. Give Random Forest sufficient attention because it is your actual benchmark; use other methods to position the study without implying that all will be implemented.

Connection to the study: Justifies the conventional benchmark and introduces the question of how models represent spatial relationships.

2.1.8 Graph Neural Networks and Temporal Modeling

- **2.1.8.1 Graph representations and neighborhood aggregation**
- **2.1.8.2 Graph convolution and GraphSAGE**
- **2.1.8.3 LSTM encoding of historical environmental features**
- **2.1.8.4 Integration of temporal encoding and graph aggregation**
- **2.1.8.5 Graph learning for temperature and environmental estimation**

Explain the components of the proposed LSTM–GraphSAGE model using recent methodological literature and applications. Clearly distinguish studies using frequent time-series observations from your two annual input steps.

Connection to the study: Provides the methodological foundation for the proposed architecture without assuming it will outperform alternatives.

2.1.9 Evaluation of Connectivity Contributions and Predictive Performance

- **2.1.9.1 Controlled ablation and connectivity-disabled comparisons**
- **2.1.9.2 Comparable predictors, training conditions, and held-out observations**
- **2.1.9.3 Spatial dependence, shared observation support, and data leakage**
- **2.1.9.4 RMSE, MAE, and R^2 for thermal estimation**
- **2.1.9.5 Uncertainty, repeated experiments, and limits of generalization**

This should be a major section. Review how studies isolate the contribution of a model component and evaluate spatial predictions fairly.

Connection to the study: Directly supports Research Questions 1–3, especially the distinction between testing the whole model against Random Forest and testing connectivity through the matched ablation.

2.1.10 Thermal Hotspot Identification and Spatial Interpretation

- **2.1.10.1 Definitions and thresholds for elevated SUHI**
- **2.1.10.2 Threshold-based hotspots and statistical spatial clustering**
- **2.1.10.3 Agreement between predicted and observed hotspot locations**
- **2.1.10.4 Mapping uncertainty and appropriate planning interpretation**

Review how hotspots are defined and evaluated. Distinguish a threshold-based definition from a formal clustering method so that the terminology matches the eventual methodology.

Connection to the study: Supports Research Question 4 and the interpretation of the resulting maps.

2.1.11 Synthesis of Related Literature

Bring the themes together around three questions:

- What does existing evidence establish about satellite-derived urban thermal estimation?
- What remains uncertain about the additional predictive value of road connectivity?
- What comparison and validation procedures are needed to investigate that uncertainty?

This should synthesize previously reviewed sources. Avoid asserting that no comparable study exists before completing the search.

2.2 Related Systems

Use this section for **specific implemented research frameworks, prototypes, and relevant mapping platforms**. Section 2.1 explains the ideas and evidence; Section 2.2 examines how selected implementations combine data, models, evaluation, and outputs.

The headings below are system categories. Once sources are verified, individual systems or published frameworks can become named subsections within them.

2.2.1 Satellite-Based Urban Thermal Mapping Systems

Review implementations that acquire satellite observations, process LST or SUHI, and produce spatial thermal outputs.

Examine their data sources, reference definitions, coverage, processing workflow, and validation.

2.2.2 Machine-Learning Thermal Estimation Systems

Review implemented regression and raster deep-learning frameworks, prioritizing those comparable to your satellite-derived target and environmental predictors.

Examine their input variables, prediction targets, spatial units, baselines, and evaluation design.

2.2.3 Graph-Based and Spatiotemporal Temperature Estimation Systems

Give the most attention to implemented graph frameworks for LST, SUHI, or road-surface temperature. Include adjacent environmental applications selectively where they clarify an architectural choice.

Examine node definitions, edge construction, temporal inputs, aggregation methods, ablation experiments, and transferability to your study.

2.2.4 Geospatial Thermal Visualization and Decision-Support Systems

Review systems that communicate temperature estimates, hotspots, and uncertainty through maps or interfaces.

Focus on features relevant to your planned outputs. A full operational dashboard or heat-warning service should not become an implied thesis requirement.

2.2.5 Comparative Analysis of Related Systems

Create a comparison matrix using consistent criteria:

Comparison criterion	What to record
Target	LST, SUHI, air temperature, or road-surface temperature
Data	Sensors, predictors, study period, and geographic setting
Spatial representation	Pixels, areas, stations, or road-network nodes
Connectivity	How edges are defined and used
Temporal component	Input frequency, sequence length, and target period
Evaluation	Baselines, ablations, partitions, and metrics
Outputs and limitations	Maps, hotspot assessment, uncertainty, and reproducibility

2.2.6 Synthesis and Positioning of the Proposed Study

Conclude by explaining how your proposed study responds to the reviewed evidence through its combination of:

- Road-associated satellite features;
- LSTM–GraphSAGE estimation;
- A matched connectivity-disabled comparison;
- A separate Random Forest benchmark;
- Reference-based SUHI calculation and held-out hotspot assessment.

Describe this as the proposed contribution to be evaluated, with the final gap statement determined by the completed review.

Suggested distribution of sources

Aim initially for **95 unique sources**, allocating each to a primary topic for tracking. A source may support several sections but counts only once.

Primary topic	Target sources
SUHI concepts and tropical/local context	8
SUHI measurement and reference selection	7

Satellite observations and data quality	10
Environmental predictors	9
Road-network representation	8
Satellite–road data integration	7
Conventional and raster machine learning	9
Graph and temporal modeling	12
Ablation and spatial validation	9
Hotspot identification and interpretation	6
Additional system-specific sources	10
Total	95

The system reviews will also reuse relevant papers introduced in Section 2.1. We should select a manageable set of well-documented systems for deeper comparison rather than treating every source as a separate system.

The greatest depth should go to Sections 2.1.5–2.1.9 and 2.2.3–2.2.5. Those sections explain and justify your computer-science contribution. Chapter 2 should establish the evidence behind methodological choices; exact buffer sizes, graph settings, model parameters, and partitioning procedures belong in Chapter 3.